

Program : Diploma in Communication and Computer Networking	
Course Code : 4321	Course Title: Introduction to Microcontrollers
Semester : 4	Credits: 4
Course Category: Programme core course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Understand basics of Microcontrollers and their applications
- Understand Architecture and Organization of Microcontrollers.
- Understand Interrupts in 8051 and serial communication.
- Understand interfacing of a microcontroller for communication.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Sound knowledge in Logic System Design and Computer organization & architecture.		Logic Design , Computer Architecture & Organization in S3	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	To explain basics of Microcontrollers and their applications.	15	Understanding
CO2	To observe the Architecture and Organization of Microcontrollers	15	Applying
CO3	To demonstrate interrupts and serial communication in 8051	15	Applying
CO4	To use microcontroller for communication	15	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	1		1			
CO2	3	2		2			
CO3	3	2		2			
CO4	3	2		2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To explain the basics of Microcontrollers and their applications.		
M1.01	To understand Microcontrollers	8	Understanding
M1.02	To understand Applications of Microcontrollers	7	Understanding
Contents: Introduction to Microcontrollers: Introduction, Comparison between microprocessors and microcontrollers, Microcontroller families, Types of Microcontrollers - Criteria for selecting a microcontroller - Example Applications.			
CO2	To observe the Architecture and Organization of Microcontrollers.		
M2.01	To understand features of 8051	7	Understanding
M2.02	To understand addressing modes and instruction set	8	Applying
	Series Test –I	1	
Contents: Architecture and organization of Microcontrollers: Microcontrollers - Characteristics and Resources of a microcontroller, 8051- features, architecture, memory organization, registers, I/O ports, pin configuration and functions. 8051 Addressing Modes, Different types of instructions and Instruction Set, Simple Assembly language programming examples for 8051			
CO3	To demonstrate interrupts and serial communication in 8051		
M3.01	To understand interrupts in 8051	8	Applying

M3.02	To understand Serial Communication	7	Applying
Contents: Interrupts and serial communication in 8051: Interrupts in 8051: Types, interrupt source, interrupt handling and programming, Timer/Counter programming: Operating modes, time delay generation, Waveform generation. Serial communication: RS 232 interface, registers in UART, modes of operation, programming examples for serial data transmission and reception.			
CO4	To use microcontroller for communication		
M 4.01	To understand interfacing with devices	15	Applying
	Series Test – II	1	
Contents: Interfacing of a microcontroller for communication: Interfacing: Interfacing (block schematic and assembly language programming) of DIP switch, stepper motor, ADC, DAC, LEDs and seven segment displays, alphanumeric LCD module with 8051			

Text / Reference

T/R	Book Title/Author
T1	Kenneth J. Ayala, The 8051 Microcontroller, Cengage learning, 3/e.
T2	Raj Kamal, <i>Microcontrollers: Architecture, Programming, Interfacing and System Design</i> , Pearson Education, 2011.
T3	Lyla B.Das : Microprocessors and Microcontrollers, Pearson Education, India, 2011
R1	Douglas V. Hall, SSSP Rao, <i>Microprocessors and Interfacing</i> , Third Edition, McGrawHill Education, 2012.
R2	Han Way Hung, “PIC Microcontroller, An introduction to software and hardware interfacing “, Cenage learning
R3	I.Scott Mackenzie, Raphel C.-W Phan, The 8051 microcontroller, 4th edition.
R4	Muhammed Ali Mazidi, The 8051 Microcontroller and Embedded Systems, Pearson Education, 2nd edition
R5	Nagoorkani, Microprocessors and Microcontrollers 2e, McGraw Hill Education India, 2012
R6	Nagoorkani, Microprocessors and Microcontrollers 2e, McGraw Hill Education India, 2012

Online Resources

Sl.No	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ee18
2	https://study.unisa.edu.au/courses/151843